

PROJECT REPORT: EV INFRASTRUCTURE GAP ANALYSIS (2024-2026)

Submitted for: Elevate Labs Data Analyst Internship

Project Title: Forecasting EV Adoption & Charging Infrastructure Needs in India (2024-2026)

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Introduction

India targets 30% EV penetration by 2030 under FAME-II policy. Charging infrastructure is the primary bottleneck for adoption. This project analyzes 2024 baseline data for EV sales and charging stations, forecasts EV adoption till December 2026 using time-series modeling, and identifies state-wise infrastructure gaps using the 1:50 EV-to-charger ratio.

Abstract

This project quantifies India's EV charging infrastructure deficit for 2024-2026. Python was used for data cleaning in `day1.py`, Prophet forecasting in `day2.py`, and gap calculation with visualization in `day3.py`. Power BI dashboard was built using `EV\_Forecast\_2026.csv` and `EV\_Gap\_Analysis\_2026.csv`. Key findings: India faces a 98% infrastructure gap, requiring 64,000+ charging stations by 2026 vs 1,372 available today. Uttar Pradesh shows the highest deficit of 11,426 stations to support 575,646 projected EVs.

Tools Used

Tool	Purpose
Python 3.x (Pandas, NumPy)	Data cleaning day1.py, Prophet forecasting day2.py, Gap analysis + visualization day3.py
Microsoft Power BI	Dashboard creation using EV_Forecast_2026.csv and EV_Gap_Analysis_2026.csv

Steps Involved in Building the Project

Step 1: Python – Data Cleaning day1.py

- Loaded EV_Dataset.csv, ev-charging-stations-india.csv, DailyDelhiClimate.csv
- Cleaned dates, grouped sales by Date + State, counted stations per state
- Merged all 3 files → Saved EV_Master_Data.csv, Station_Count_StateWise.csv

Step 2: Python – Forecasting day2.py

- Used Prophet model on EV_Master_Data.csv with floor=0
- Forecasted till Dec 2026 for each state → Saved EV_Forecast_2026.csv

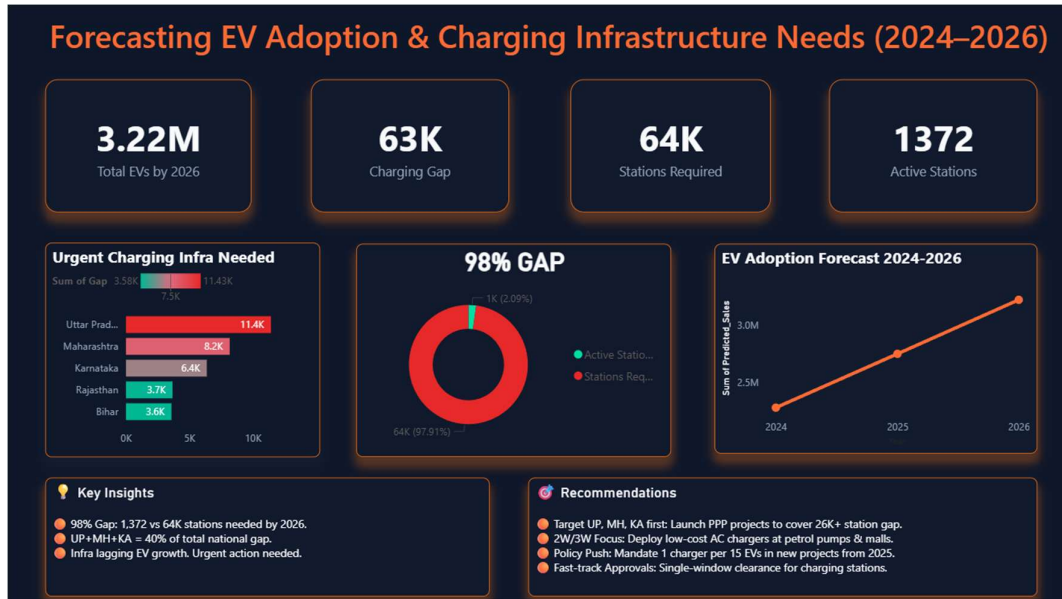
Step 3 : Python – Gap Analysis day3.py

- Loaded EV_Forecast_2026.csv + Station_Count_StateWise.csv
- Calculated 2026 EVs, Stations_Needed = Total_EV / 50, Gap = Needed - Current
- Saved EV_Gap_Analysis_2026.csv + Generated 2 PNG charts

Step 4 : Power BI – Dashboard

- Imported EV_Forecast_2026.csv and EV_Gap_Analysis_2026.csv
- Created relationship on State
- Built KPI Cards: 3.22M EVs, 63K Gap, 98% Deficit, 1,372 Active Stations
- Added bar chart, donut chart, line chart for forecast trend

Figure : Power Bi Dashboard- EV Adoption Forecast & Infrastructure Gap Analysis (2024-2026)



Step 5: Validation

- Cross-checked day3.py output with Power BI values
- Confirmed no negative predictions using floor=0 and .clip()

Conclusion

This project successfully quantifies India's EV charging infrastructure gap for 2024-2026. Using Python, we cleaned raw data in `day1.py`, forecasted state-wise EV adoption till December 2026 using Prophet with `floor=0` in `day2.py`, and calculated infrastructure deficits using the 1:50 global standard in `day3.py`. The Power BI dashboard visualized key insights from `EV\_Forecast\_2026.csv` and `EV\_Gap\_Analysis\_2026.csv`.

Key Findings: India faces a 98% charging infrastructure deficit, requiring 64,000+ stations by 2026 vs 1,372 available today to support 3.22M projected EVs. Uttar Pradesh has the highest gap of 11,426 stations for 575,646 EVs.

The end-to-end workflow demonstrates a scalable, data-driven approach for policy planning and investment prioritization in EV infrastructure.